Diagnostic Features

- Presence of multilayered epidermis with thick cuticle.
- Sunken stomata are present only in the lower epidermis.
- Mesophyll is well differentiated into palisade and spongy parenchyma.
- Mechanical tissues are well developed.

II - Fresh or preserved specimens and Models / Photographs / Charts

Exercise 4: Adaptations of flowers for pollination by different agents.

Aim: To study the adaptations in flowers for pollination by different agents (wind and insects)

Principle: The process of transfer of pollen grains from the anther to stigma of a flower is called **pollination**.

Requirements: Fresh flowers of maize or any other cereal / gram, any insect pollination flowers like *Salvia*, *Calotropis, Ocimum* and Asteraceae flowers.

Place the given flower on a slide and observe it with the help of hand lens. Note down the adaptations of the flowers meant for pollination by the external agents.

5 A. Wind Pollinated Flowers - Anemophily

Diagnostic Features

- The flowers are small, inconspicuous, colourless, odourless and nectarless.
- Anthers and stigmas are commonly exerted.
- Pollen grains are light, small, powdery and produced in large numbers.
- The stigmas are large, sometimes feathery and branched adapted to catch the pollens.

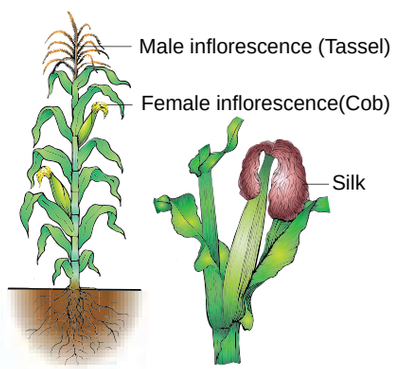


Figure 4a: Maize

5 B. Insect Pollinated Flowers - Entemophily

Diagnostic Features

- The flowers are showy, brightly coloured and scented.
- The flowers produce nectar or edible pollen.
- Anthers and stigmas are commonly inserted.
- Stigmas are usually unbranched and flat or lobed.

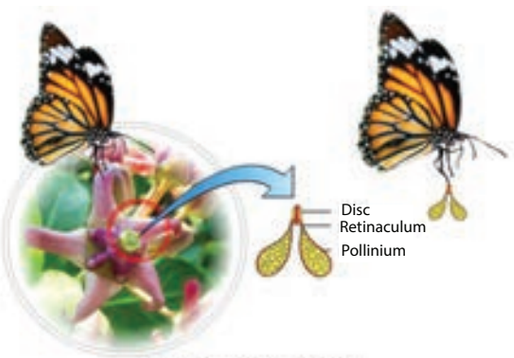


Figure 4b: *Calotropis*

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